

MY PLACEMENT WITH THE CENTRAL BANK

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I completed my 12-week training placement with the Central Bank of Ireland in September of this year. Over this period, I worked on a project titled 'Unwrapping Bond Secrets with AI'. I'm sure that many of you have been affected by Covid-19 in some form or another, so you won't be surprised to hear that the bond markets were no exception. The global pandemic was a shock to the financial system. The uncertainty generated by these sorts of shocks can reduce the liquidity of the market because market participants may lower their risk appetite. A bond's liquidity refers to the ease with which a bond can be converted into cash, i.e. sold, without affecting its market price.



For those of you unfamiliar with markets and liquidity, consider the following example: Under periods of 'normality' a specific bond has a market value of €10k, however, the global markets undergo a stressful period, and the number of buyers is reduced. Some owners of this bond are forced to sell due to the market conditions, with a lack of buyers they are forced to reduce the price from €10k to €5k, making a loss on each bond sold. The losses experienced could pose a threat to the long-term viability of the bond owners' business. There is of course a risk that if one of these companies fails, the knock-on effect in the markets could lead others to fail. The reason this is of concern to the Central Bank is because bond owners could be insurance companies, or pension funds, or commercial banks. Ultimately, a liquidity crisis could directly or indirectly have negative consequences for the public.

The Central Bank has a responsibility to ensure the stability of the financial system, protect consumers of financial services and regulate financial institutions. The regulatory power of the Central Bank allows it to monitor the bond portfolios of certain companies operating in Ireland. Thus, my project produced a way of assessing the liquidity of bonds held by the aforementioned bond owners. Unfortunately, liquidity is not something one can measure. Instead, we can observe the bid-ask price spread, or the number of trades in a given time period, or the number of buyers and sellers in the market, and try to gauge liquidity from these variables.

We chose to contain our analysis to the bid-ask price spread, which is the difference between the price you can buy and sell a bond at, since it is considered as the best indicator of liquidity. Using daily bid-ask price spreads for hundreds of thousands of bonds over a 5-year period I designed an algorithm to predict the future bid-ask price spread. The algorithm first clustered the bonds into five clusters using the DBA-K-means algorithm. LSTM and ARIMA models were then used to train and predict the bid-ask price spread for each bond in each cluster. If the difference between the observed bid-ask price spread and the predicted bid-ask price spread was above a specified threshold then it was considered an anomaly. Bonds that suffer many anomalies are worth investigation by supervisors at the Central Bank as it may indicate a larger liquidity risk.

I enjoyed working on a significant project which can offer great impact to the Central Bank of Ireland and the public. The work done can hopefully assist with the regulation of financial services firms in Ireland. I'd like to thank everyone at the Central Bank for their help throughout the project, in particular Kate Rooney, Eugene Kashdan and Davide Cellai.